

INDUSTRY-STANDARD CURRICULUM

Frontend Developer

Zero to Hero

A structured 3-month program from complete beginner to job-ready frontend developer.

Instructor: Paul Olagboye

3 Months

Program

72 Hours

Total

12 Weeks

36 Sessions

3 Projects

Capstone



Web Development

Three Months. One Transformation.

Month 01

Foundations

HTML5, CSS3, Flexbox,
Grid, Responsive Design

Weeks 1–4

Goal: Responsive portfolio
deployed on GitHub Pages

Month 02

Core Dev Skills

JavaScript ES6+,
DOM, APIs, Git/GitHub

Weeks 5–8

Goal: Interactive app
deployed on Netlify

Month 03

React & Deploy

React, Hooks, Router,
Tailwind CSS, Vercel

Weeks 9–12

Goal: Full React app
on Vercel

MONTH 01

Web Foundations & HTML/CSS Mastery

Weeks 1–4 · 12 sessions

W1: How the Web Works & Your First Webpage

W2: CSS Deep Dive — Layouts with Flexbox

W3: CSS Grid & Responsive Design

W4: Portfolio Capstone — Month 1 Project

01

How the Web Works & Your First Webpage

Environment setup · HTML5 basics · Dev tools

Session 1

The Web & Your Environment

- How browsers, servers & the internet work
- Installing VS Code, extensions & dev tools
- Anatomy of an HTML document (DOCTYPE, html, head, body)
- Your first webpage: headings, paragraphs, links, images

Session 2

HTML Structure & Semantic Elements

- Semantic HTML5: header, nav, main, section, article, footer
- Lists, tables, forms & input types
- HTML attributes: id, class, href, src, alt
- Validating HTML with W3C validator

Session 3

Intro to CSS — Selectors & Box Model

- Linking CSS to HTML (external stylesheet)
- Selectors: element, class, id, descendant
- The Box Model: margin, border, padding, content
- Colors, fonts, background-color, text properties

 **Week 1 Project:** Build a styled "About Me" webpage with a bio, photo, and links to your social profiles.

CSS Deep Dive — Layouts with Flexbox

Flexbox · Typography · CSS variables

Session 4

Typography & Visual Styling

- Google Fonts integration — font pairing principles
- font-size, line-height, letter-spacing, text-align
- Colors: hex, rgb, hsl, opacity
- CSS custom properties (variables) for theming

Session 5

Flexbox Layout System

- display: flex — main axis vs cross axis
- justify-content, align-items, align-self
- flex-grow, flex-shrink, flex-basis
- Building navigation bars and card rows with Flexbox

Session 6

Flexbox Project & Pseudo-classes

- hover, focus, active, nth-child pseudo-classes
- Transitions: ease, duration, property
- Building a product card component
- Position: relative, absolute, sticky, fixed



Week 2 Project:

Design a responsive product card grid (4+ cards) with hover effects and a sticky navigation bar.

CSS Grid & Responsive Design

CSS Grid · Media queries · Mobile-first

Session 7

CSS Grid System

- display: grid — rows, columns, gaps
- grid-template-columns: repeat, fr units
- grid-column / grid-row spanning
- Named grid areas: grid-template-areas

Session 8

Responsive Design & Media Queries

- Mobile-first design philosophy
- Media queries: min-width breakpoints (480, 768, 1024)
- Responsive images: max-width: 100%, object-fit
- Viewport meta tag, dvh, svh units

Session 9

CSS Animations & Figma Intro

- @keyframes, animation-duration, iteration-count, fill-mode
- Transform: translate, scale, rotate
- Intro to Figma: inspecting designs, copying CSS values
- prefers-reduced-motion accessibility media query



Week 3 Project:

Build a responsive blog layout using CSS Grid — desktop 3-column, tablet 2-column, mobile 1-column.

Portfolio Capstone — Month 1 Project

Project week · Code review · Deployment basics

Session 10

Portfolio Planning & Structure

- Information architecture — what makes a great portfolio
- Wireframing your sections: Hero, About, Projects, Contact
- Writing semantic HTML for the portfolio
- Integrating Google Fonts & custom CSS properties

Session 11

Portfolio Styling & Responsiveness

- CSS Grid for project section layout
- Flexbox for navigation and footer
- Animations: scroll-reveal effects with CSS
- Responsive across all breakpoints — mobile through desktop

Session 12

Code Review & GitHub Pages Deploy

- Introduction to GitHub — creating a repo, pushing code
- GitHub Pages: deploying a static site
- Accessibility basics: alt text, color contrast, keyboard nav
- Peer code review session — giving and receiving feedback



MONTH 02

JavaScript & Developer Workflow

02

Weeks 5–8 · 12 sessions

W5: JavaScript Foundations

W6: The DOM — Bringing Pages to Life

W7: Async JavaScript & APIs

W8: Git, GitHub & Professional Workflow

JavaScript Foundations

Variables · Data types · Functions · Control flow

Session 13

JS Basics & Terminal/Git Setup

- Intro to terminal: cd, ls, mkdir, touch commands
- Git init, add, commit, push — the daily workflow
- Variables: var, let, const — when to use which
- Primitive types: string, number, boolean, null, undefined

Session 14

Functions & Control Flow

- Function declarations vs arrow functions
- Parameters, arguments, return values
- if / else if / else, ternary operator
- switch statements, truthy/falsy values

Session 15

Arrays, Objects & Loops

- Array methods: push, pop, map, filter, find, reduce
- Object literals: properties, methods, destructuring
- for...of, forEach, for...in loops
- Spread operator and rest parameters



Week 5 Project:

Build a JavaScript quiz app — hardcoded questions, score tracking, show/hide correct answers.

The DOM — Bringing Pages to Life

DOM manipulation · Events · Forms

Session 16

DOM Selection & Manipulation

- `querySelector`, `querySelectorAll`, `getElementById`
- `textContent`, `innerHTML`, `value`
- `classList`: `add`, `remove`, `toggle`, `contains`
- `createElement`, `appendChild`, `removeChild`, `insertAdjacentHTML`

Session 17

Events & Event Listeners

- `addEventListener`: `click`, `input`, `keydown`, `submit`
- Event object: `target`, `preventDefault`, `stopPropagation`
- Event delegation — handling dynamic elements
- Building interactive UI: tabs, accordions, toggles

Session 18

Forms, Validation & Local Storage

- HTML5 form validation attributes
- Custom JavaScript validation with error messages
- `localStorage`: `setItem`, `getItem`, `removeItem`, `JSON.stringify`
- Building a persistent to-do list with local storage



Week 6 Project:

Build a full To-Do List app with add, complete, delete, and filter tasks — persisted in `localStorage`.

Async JavaScript & APIs

Promises · Fetch · async/await · REST APIs

Session 19

Asynchronous JavaScript & Promises

- The call stack, event loop, and callback queue
- Callback hell — why we needed something better
- Promises: `.then()`, `.catch()`, `.finally()`
- `Promise.all` and `Promise.race` for parallel requests

Session 20

Fetch API & REST Principles

- HTTP verbs: GET, POST, PUT, DELETE
- JSON: `parse` and `stringify`
- `fetch()` with `async/await` — clean, readable code
- Handling errors: `try/catch`, HTTP status codes

Session 21

API Integration Project Setup

- Reading API documentation (OpenWeatherMap, TMDb)
- Query parameters and API keys — `.env` basics
- Rendering API data to the DOM dynamically
- Loading states, error states, and empty states



Week 7 Project:

Weather or Movie Search app that fetches live data from a public API and renders results dynamically.

Git, GitHub & Professional Workflow

Branching · PRs · Code quality · ES Modules

Session 22

Git Branching & Collaboration

- git branch, checkout, merge, rebase basics
- Pull requests on GitHub — the real-world workflow
- Resolving merge conflicts — practical exercises
- GitHub README files: markdown, badges, project descriptions

Session 23

ES Modules & Code Organisation

- import / export: named and default
- Separating code into multiple files — clean architecture
- Introduction to npm and package.json
- Using a third-party library (e.g., date-fns, axios)

Session 24

Month 2 Project Polish & Review

- Code review: clean naming, DRY principles
- Performance: debounce, throttle for event handlers
- Accessibility: keyboard navigation, ARIA live regions
- Deploying to Netlify — drag and drop + CLI deploy



Month 2 Capstone:

Polished interactive app deployed on Netlify with GitHub repo, README, and clean commits.

MONTH 03

React, State Management & Deployment

Weeks 9–12 · 12 sessions

W9: React Fundamentals

W10: React Hooks & API Integration

W11: React Router & Tailwind CSS

W12: Capstone Project & Career Readiness

03

React Fundamentals

Components · JSX · Props · Vite setup

Session 25

Why React & Project Setup with Vite

- The component model — why React dominates the industry
- Create a React app with Vite (not Create React App)
- JSX syntax — writing HTML-like code in JavaScript
- Rendering components, nesting, and the React tree

Session 26

Components, Props & Rendering Lists

- Functional components — the modern React standard
- Props: passing data parent-to-child
- PropTypes for basic type checking
- Rendering lists with `.map()` and the key prop

Session 27

State with `useState` Hook

- What is state? Local vs derived vs server state
- `useState`: reading and updating state values
- Controlled components — inputs driven by state
- Conditional rendering: `&&`, ternary, early return

React Hooks & API Integration

useEffect · useContext · Custom hooks · Fetch in React

Session 28

useEffect & Data Fetching

- useEffect: side effects, dependency array, cleanup
- Fetching API data inside useEffect — the pattern
- Loading, error, and success states in React
- Async functions inside useEffect (the wrapper pattern)

Session 29

useContext & Global State

- Prop drilling problem and why context solves it
- createContext, Provider, useContext pattern
- Building a theme switcher (light/dark) with context
- When to use context vs local state vs external store

Session 30

Custom Hooks & Code Reuse

- Extracting logic into custom hooks: useFetch, useLocalStorage
- Rules of hooks — why they exist and what they enforce
- useRef: DOM access, persist values without re-render
- React Developer Tools: profiling and debugging



Week 10 Project:

Rebuild your Month 2 API app in React with useFetch custom hook, loading states, and theme toggle.

React Router & Tailwind CSS

Multi-page apps · Routing · Tailwind utility-first CSS

Session 31

React Router v6

- Installing and configuring React Router DOM v6
- BrowserRouter, Routes, Route, Link, NavLink
- useParams, useNavigate, useLocation hooks
- Protected routes pattern — auth-gated pages

Session 32

Tailwind CSS with React

- Utility-first CSS philosophy — why it works
- Installing Tailwind with Vite + PostCSS config
- Responsive utilities: sm:, md:, lg: prefixes
- Dark mode: dark: prefix and class strategy

Session 33

Forms, Validation & Component Libraries

- Controlled vs uncontrolled forms in React
- Basic form validation with React state
- Introduction to shadcn/ui — copy-paste components
- Building a multi-page layout with persistent navigation



Week 11 Project:

Multi-page React app (e.g., a blog or recipe site) with React Router and Tailwind CSS styling.

Capstone Project & Career Readiness

Final project · Deployment · Interview prep · Next steps

Session 34

Capstone Project: Build

- Students build their chosen capstone: e-commerce store, task manager, portfolio v2, or social feed
- Instructor 1-on-1 reviews and code guidance
- Performance: React.memo, lazy loading, code splitting
- Vercel deployment: CI/CD pipeline from GitHub

Session 35

Capstone Project: Polish & Deploy

- SEO basics: meta tags, Open Graph, title/description
- Web accessibility audit with Lighthouse
- Environment variables in Vite + Vercel
- Writing a professional README with screenshots and demo link

Session 36

Career Readiness & Graduation

- Portfolio presentation: how to talk through your projects
- Common frontend interview questions (live coding review)
- What to learn next: TypeScript, Next.js, Testing, Node.js
- Industry resources: roadmap.sh, Frontend Mentor, MDN docs



Final Capstone:

Deployed, polished React application on Vercel with GitHub repo, README, and live demo — ready for your portfolio.

Industry-standard tools introduced progressively



VS Code

Week 1



Browser DevTools

Week 1



Figma (Basics)

Week 3



Git & GitHub

Week 5



Terminal / CLI

Week 5



npm / Node.js

Week 7



Vite

Week 9



React

Week 9



Tailwind CSS

Week 11



Vercel / Netlify

Week 12

What you'll be able to do



Build responsive, accessible websites from scratch

Using semantic HTML5 and modern CSS including Flexbox, Grid, custom properties, and animations.



Write clean, modern JavaScript (ES6+)

DOM manipulation, event handling, async/await, fetch API, and working with real third-party APIs.



Work professionally with Git & GitHub

Version control, branching, pull requests, and collaborative workflows — the way real teams work.



Build dynamic React applications

Components, props, state, hooks (useState, useEffect, useContext), React Router, and API integration.



Deploy and share real projects

3 portfolio-ready projects deployed to Vercel or Netlify — demonstrable skills for job applications.

START BUILDING TODAY

Your Coding Journey Starts Here

72 Hours

of instruction

36 Sessions

structured learning

3 Projects

portfolio-ready

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